

Course Code : ECT 253

KQLR/RS – 24 / 1545

**Third Semester B. Tech. (Electronics and Communication Engineering)
Examination**

SIGNALS AND SYSTEMS

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory and carry marks as indicated.
- (2) Assume suitable data wherever necessary.
- (3) Illustrate your answers wherever necessary with the help of neat sketches.
- (4) Due credit will be given to neatness.
- (5) Use of Non – programmable calculator is permitted.

1. (a) Sketch the following Signals :

(i) $x(t) = r(t) - 2r(t - 1) + r(t - 2)$

(ii) $x(t) = r(t) - r(t - 1) - u(t - 1) + r(t - 2) - r(t - 3) - u(t - 3)$

(iii) $x(n) = u(-n + 2) u(n)$ 6(CO1)

(b) Compute the RMS value of the following signals :

(i) $x(t) = e^{j\alpha t} \cos(\omega t)$

(ii) $x(t) = Ae^{j\omega t}$ 4(CO1)

2. (a) The system is represented as below :

$$y(n) = x(n)u(n)$$

Check the system for :

(i) Linear or Non-Linear.

(ii) Causal or Non-Causal.

(iii) Time Invariant or Time Variant. 6(CO1)

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Contd.

- (b) Elaborate the following properties of the system with example :
- (i) Stability,
 - (ii) Causality,
 - (iii) Linearity,
 - (iv) Time-invariance. 4(CO1,4)
3. (a) The input and impulse response of DT-LTI system are
 $x[n] = u[n] - 2u[n - 2] + u[n - 5]$;
 $h[n] = \delta[n + 2] + 2\delta[n + 1] + 3\delta[n - 1] + 4\delta[n - 2] + 5\delta[n - 3]$ respectively.
 Obtain the response $y[n]$ of the system. 6(CO3)
- (b) The impulse responses of the LTI systems are given below. Verify the causality and stability of the system :
- (i) $h_1[n] = [3]^n u(2 - n)$
 - (ii) $h_2[t] = e^{3t} u(-2 - t)$ 4(CO3,4)
4. (a) CT-LTI system is described as –

$$\frac{d^2y(t)}{dt^2} + 7\frac{dy(t)}{dt} + 6y(t) = 7x(t) + 11\frac{dx(t)}{dt}$$
 Determine the impulse response $h(t)$ if the system is both stable and causal. 6(CO2)
- (b) Obtain the impulse response of the CT-LTI stable and causal system whose input is $x(t) = e^{-0.5t}$ and the output is $y(t) = e^{-t} - 2e^{-2t} + e^{-3t}$. 4(CO2)
5. (a) Determine $H(z)$ for the system described by following difference equation and hence find its impulse response. Comment on the stability of the system.
 $8y[n] - 6y[n - 1] + y[n - 2] = x[n] + x[n - 1]$ 6(CO3)
- (b) Obtain the inverse z-transform of $X(z) = \frac{z}{(z + 2)(z - 3)}$ for the ROC $|z| < 2$. 4(CO3)

6. (a) A causal LTI system is represented by the differential equation

$$\frac{d^2y(t)}{dt^2} + 6 \frac{dy(t)}{dt} + 8y(t) = 2x(t).$$

Find the impulse response of the system using CTFT. Plot the magnitude and phase spectrum of this system. 6(CO5)

- (b) A DT LTI system is represented by

$$H(e^{j\omega}) = \frac{1}{1 + \frac{1}{2} e^{-j\omega} - \frac{1}{8} e^{-j2\omega}}.$$

Deduce the difference equation representing the input-output relationship of a DT system.

OR

Determine the Fourier Series representation for the signal :

$$x(t) = 4 \cos \left(\frac{\pi}{2} t + \frac{\pi}{4} \right)$$

Plot its magnitude and phase spectrum. 4(CO5)

